Standalone Apps vs. Social Media: A Participatory Evaluation of Digital Maternal Health Delivery in Rural Limpopo, South Africa

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Abstract

Maternal malnutrition in rural Limpopo is worsened by poor connectivity, high data costs, and a lack of culturally relevant guidance, leaving pregnant women and their support structures without effective access to nutritional information. We conducted a three-day, three-site participatory evaluation with pregnant women, health workers, and decision-makers, comparing AMEN, an offline-first maternal health app, against equivalent content on TikTok and YouTube, using a usability scale, co-design activities, interviews, and field notes analyzed through reflexive thematic analysis. Findings show no single channel meets this community's needs on its own: digital habits, literacy, and access constraints vary too widely across groups, and while AMEN was trusted as a reliable repository of clinical content, a persistent gap between that trust and its interface usability points to social media as the higher-reach channel needed to drive users toward it. These results point to a cooperative multi-channel model for maternal health delivery and offer transferable design recommendations for similar low-resource, rural contexts.

Keywords

Maternal Health, mHealth, Multi-Channel Content Delivery, Participatory Co-Design, Low-Literacy User Research, Rural South Africa, Offline-First Design

1 Introduction

In rural Limpopo, mothers often lack basic nutritional knowledge, and this contributes to widespread under-nutrition. Part of the problem is that standard dietary advice often conflicts with local cultural practices, which makes it less effective [5, 15]. This is why the AMEN project, a collaboration between University College Dublin (UCD) and the University of Limpopo (UL), was set up. It aims to use digital health tools designed for the local context to fight malnutrition, support agriculture, and improve nutrition education for pregnant women and the people who support them in rural Limpopo. The project team has already built up a base of nutritional knowledge rooted in local foods, culture, and trusted community voices.

So the real question was not what information to share, but how to share it. A mobile app seemed like the obvious answer at first. But as the project developed, a bigger question came up: which channels actually work best for reaching pregnant women and their support networks with this kind of content, in a way that is usable, trusted, and culturally appropriate? And what can this tell us about designing digital health tools for low-resource settings more broadly? Formally, this study asks: which combination of digital channels, a dedicated offline-first app, TikTok, and YouTube, best meets these usability, trust, and accessibility needs, and what does this reveal about digital health delivery design in low-resource settings?

To explore this, a fieldwork study was run in Limpopo. Health workers, decision-makers, and pregnant women interacted with a mobile app prototype and two social media channels. This paper uses reflexive thematic analysis to look at what was learned, drawing on multiple sources of data. The findings point to what people prefer across different digital channels, offer guidelines for designing mobile apps in this context, and share lessons from running co-design workshops. Together, they help build a clearer picture of how to design and deliver digital maternal health content in rural, low-resource settings.

2 Background and Related Work

In South Africa, MomConnect proved there is strong demand for digital maternal health support, signing up over 494,000 women in its first year [2]. But because it relies on USSD and WhatsApp, its interactive features are limited [14]. Richer apps like Pregnancy+ offer more content, but high data costs, storage demands, and poor connectivity leave most rural users excluded [13]. Users in these settings also tend to prefer audio and video over text, meaning any solution needs to balance multimedia content with offline-first design [13].

Trust in digital health content is shaped by existing social relationships rather than the technology itself. Pregnant women in these communities regularly weigh digital health advice against guidance from elders and traditional healers. Trust in digital content tends to build slowly, through relatable stories and peer testimonials rather than clinical authority alone [8, 9].

Beyond dedicated apps, social media is emerging as a primary health information channel in its own right. Most formal mHealth tools rely on dedicated, closed apps, but rural South African communities are increasingly turning to social media platforms like YouTube, TikTok, and WhatsApp as their main source of health information [11]. These platforms offer strong engagement, peer support, and navigation that already feels familiar, unlike new custom software [11]. While there is plenty of research on app-based interventions and on open social media use separately, little work has directly compared how rural users experience trust, usability, and content preferences across a dedicated offline-first app versus mainstream social platforms carrying the same localized content.

Co-design methodology offers one route to overcoming these literacy and trust barriers. Overcoming literacy and trust

barriers calls for co-design from the start [6]. Past workshops show that rural communities are open to digital health tools when these are built around local languages and culture [6]. Automated translation isn't enough here, since it doesn't handle South African languages well, so native speakers need to be involved in creating content and tools from day one [6].

The choice of design tools used during co-design is equally important in low-literacy settings. Choosing the right design tools also matters in low-literacy settings [17]. Abstract brainstorming often leaves participants facing a "blank page," unsure where to start [17]. Physical design cards work better, giving non-literate participants something hands-on to prioritize features around [17].

Evaluating these tools once built requires its own set of adapted methods. Evaluating mHealth tools in low-literacy settings also calls for methods beyond standard digital metrics [1, 12]. In one South African study, a user-centred redesign raised initial adoption from 41.2% to 72.4% and 90-day retention from 29.7% to 51.8% [12]. Usability testing needs to be adapted too: low-fidelity wireframes and paper prototypes tend to confuse users unfamiliar with digital mockups, so working, high-fidelity prototypes work far better [17]. Step-by-step technical instructions can also create barriers, whereas story-based scenarios let participants test features around real health goals [1]. While 5- or 7-point Likert scales give better statistical variance, research notes that extra choices can increase decision fatigue for respondents [10]. Because of this, UL team research in similar low-resource settings found that using simple 3-point visual scales works better to cut down on fatigue and get more genuine answers. Finally, combining quantitative task data with qualitative feedback gives a fuller picture of usability [1], and Braun and Clarke's reflexive thematic analysis offers a clear way to draw out core themes from this kind of varied, rich data [3, 4].

3 Research Design and Methodology

3.1 Research Approach

The research consisted of fieldwork over three days at three rural clinics in Limpopo. The workshops were structured as a two-part interaction plan designed to capture both immediate usability data and design preferences for three stakeholder groups: Pregnant Women, Community Health Workers (CHWs), and Decision Makers.

The sessions were organized into two parts: Part A, Predefined-Task Interaction and Channel Evaluations, and Part B, Co-Design.

3.2 Positionality and Reflexivity

Conducting this research involved navigating distinct insider and outsider dynamics. As a computer science honours student focused on software development and human-computer interaction rather than social science, attention was naturally drawn to technical problems and interface design rather than deep cultural details. Having no prior rural fieldwork experience, lacking Sepedi language skills, and having no lived experience in rural Limpopo also created a real risk of social desirability bias. Participants might sometimes have agreed just to be polite, especially given language gaps and power differences. Having local university team members involved helped bridge this divide significantly.

To keep these blind spots in check, the study relied heavily on triangulation. Combining field notes from multiple facilitators, input from the local team, and insights across different data sources helped ground the analysis in multiple perspectives. The analysis itself stayed closely tied to the raw data, building codes organically and grouping them into themes only after clear patterns emerged. While interpretation is always part of qualitative work, this structured approach kept the process transparent. Ultimately, these findings represent a careful interpretation shaped by an awareness of background limitations and researcher perspective.

3.3 Ethical Considerations

Due to the cultural and power-sensitive context of this fieldwork, we grounded our approach in the TRUST Code, based on the San Code of Research Ethics, which sets out four core values outlined in bold below [16]. We also followed the minimum ethical standards for ICT4D research, which asks researchers in low-resource, cross-cultural settings to stay attentive to power imbalances, defer to local collaborators, and ensure the research benefits the communities involved [7]. These principles shaped the following decisions.

Fairness shaped consent and session structure. Forms were read aloud in English and Sepedi so literacy would not determine who could take part, and sessions were split into groups of pregnant women, health workers, and decision-makers to reduce power dynamics and let everyone speak freely.

Respect meant treating participants as partners. We relied on the UL team's existing, trusted relationships with these communities, and each group session paired a Sepedi-speaking UL member with a UCT member wherever possible, giving participants a familiar facilitator.

Care shaped participant support. Participants were given a snack break between Part A and Part B, and lunch at the end of each session, as compensation for their time. We also set up mobile hotspots so the app and social channels could be used during sessions without drawing on the participants' own data. All the data collected was anonymized, and storage compliant with POPIA.

Honesty meant being upfront about the study and what participants were receiving, including that the full application was still under development.

This research received ethical clearance from the Turfloop Research Ethics Committee (TREC) at the University of Limpopo, as part of the ongoing AMEN project, and from the Inter-Faculty Research Ethics Committee (IF-REC) at the University of Cape Town, giving it both institutional and provincial approval.

3.4 Research Context and Sites

Fieldwork took place over three consecutive days across three rural clinics in the Capricorn district of Limpopo: Evelyn Lekganyane Clinic on Day 1, Makhanye Clinic on Day 2, and Ga-Dikgale Clinic on Day 3. The University of Limpopo already had established relationships with these clinics, which lowered entry barriers and helped build trust with participants from the outset (Figure 1).



Figure 1: Co-facilitating fieldwork evaluations and mobile application testing on-site with community stakeholders.

3.5 Participants

The study involved 51 participants in total: 21 health workers, 16 decision-makers (mostly grandmothers), and 14 pregnant women.

These stakeholder groups were chosen based on prior research showing how much authority figures and trusted voices shape maternal health outcomes. In rural Limpopo, a pregnant woman's nutritional choices are shaped by household hierarchies and the guidance of health workers, and maternal care is generally seen as a shared, communal responsibility. It was therefore important to include the wider support structures that influence what ends up on the household plate, not just the mothers themselves.

Recruitment differed by group. Pregnant women and health workers were recruited on the day, based on who was present at the clinic and willing to take part. Clinics knew in advance that we were coming, and fieldwork days were scheduled to align with each clinic's antenatal visit days, so pregnant women would be at the clinic to take part. Decision-makers, by contrast, were already engaged with the broader study through the UL team's prior research, and were invited by the UL team to join the workshop at each site.

3.6 Research Materials: Social Media Channels and AMEN

The UCT team built a lightweight, offline-first Android high-fidelity prototype that was installed onto participants devices where possible. It followed a layout and structure that aligned with the design of WhatsApp, as it was identified by the AMEN team as one of the main communication channels for the participants. This prototype, namely AMEN, was developed to not only act as an exposure tool for the participants, but also as a foundation for the software development team to iterate on after the handover of the codebase and research findings from the fieldwork (Figure 2).

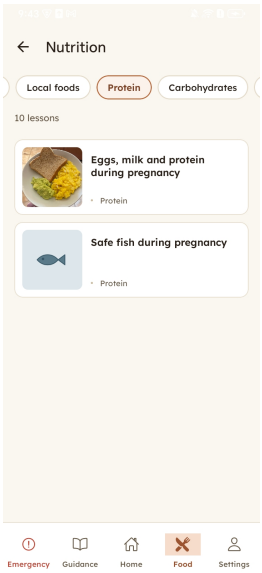
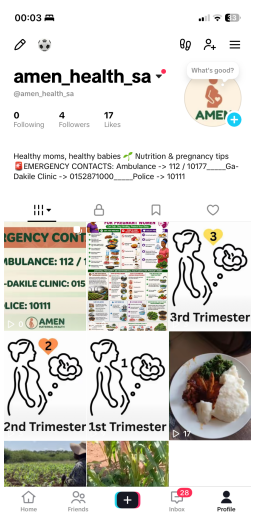
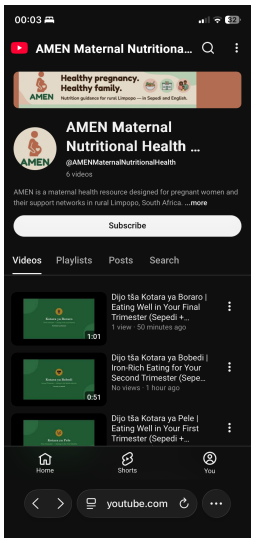


Figure 2: The dedicated offline-first AMEN mobile application interface, displaying nutrition category filter chips and lesson modules in a custom folder structure.

For this evaluation, AMEN was compared against dedicated YouTube and TikTok channels containing the equivalent trial content (Figure 3). These social media platforms served as comparative delivery channels to investigate whether a dedicated application, or platforms that the UL team had identified as part of the participants' everyday lives, better serves the usability and accessibility needs of the users.



(a) The equivalent AMEN content deployed on TikTok.



(b) The equivalent AMEN content deployed on YouTube.

Figure 3: Social media delivery channels used for comparative evaluation alongside the custom AMEN application.

All three channels acted as probing tools to provide a base from which participants could expand during the co-design activities.

3.7 Educational Trial Content and Design Constraints

The UCT team created content directly from clinical guidelines using mostly a minimalist icon and text approach. Equivalent educational material was deployed across the AMEN application, TikTok, and YouTube in three formats: a localized video encouraging food security, Sepedi audio guides on nutrition, and bilingual slideshows with icons and English and Sepedi text. The majority of the content was slideshows which created a more text-heavy interface than initially planned, but it guaranteed clinical safety, cultural respect, and factual accuracy under the time and resource constraints.

3.8 Data Collection Instruments and Procedures

The user interaction plan was comprised of two parts, and audio recordings and field notes were captured throughout.

3.8.1 Part A: Predefined-Task Interaction and Channel Evaluations. **Predefined Task Interaction:** Five identical scenario-based tasks were administered to enable direct comparison across the three channels (full task wording provided in Supplementary Information, Appendix A). The tasks were framed as real daily scenarios rather than navigation instructions, for example, finding a recipe to cook for supper. The purpose of these tasks was to obtain behavioural navigation observations to understand how users interact with the different digital interfaces. Researchers observed and documented whether participants found content easily and how they navigated the interface layout. The task-based methodology on a working prototype is supported by Till et al.'s finding that interactive, working prototypes yield more meaningful usability data than static wireframes when working with users in low-literacy, resource-constrained contexts [17].

Likert Usability and Satisfaction Scale: A 10-item scale scored as 1 (disagree), 2 (neutral), and 3 (agree) was administered for each of the channels, covering ease of use, navigation, visual clarity, independence, cultural relevance, trustworthiness, usefulness, and overall enjoyment (full item list in Supplementary Information, Appendix B). The scale was administered verbally for those who preferred or required it over reading and completing it themselves, to remove literacy barriers from the instrument itself. The decision to use a 3-point scale over the traditional 5-point scale was made to reduce cognitive load for low-literacy participants, following recommendations in the literature for simplified response formats as well as advice from the UL research team drawn from their past research in this context.

Semi-Structured Open-Ended Questions: Individual semi-structured questions followed the channel interaction and surveys, structured around three focus areas: comparative usability, digital familiarity, and socio-economic constraints (full question wording provided in Supplementary Information, Appendix C). The questions were written and read aloud in Sepedi and English, and participants could answer aloud or ask the facilitator to scribe.

3.8.2 Part B: Co-Design. After a snack break, each participant group took part in a one-hour co-design session, facilitated in English and Sepedi. Low-fidelity physical materials, cards, templates, and art supplies, were used to avoid the "blank page problem" and reduce literacy barriers (pictured in Appendix D). The five structured activities moved participants from passive evaluators to active design partners (Figure 4).

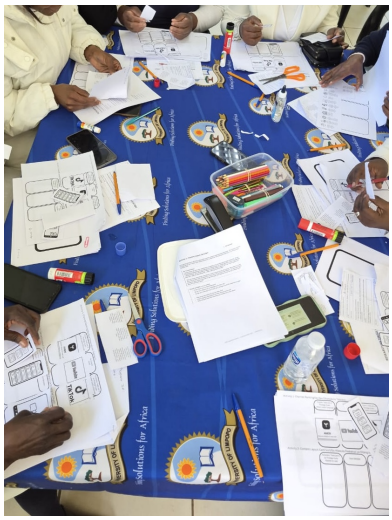


Figure 4: Participants engaging in participatory design and co-design card sorting activities during rural Limpopo fieldwork workshops.

Activity 1: Channel Ranking. Participants placed three cards, one for each channel, in order of which channel they would be most likely to use to obtain the AMEN content from. This provided a direct preference ranking per participant that could be compared across groups, field notes, and Likert scores to identify convergence or divergence between measured usability and expressed preference.

Activity 2: Content Layout Ranking. Participants ranked three content organizational layout types in order of what they find easiest to use: folder structure, search bar, and scrollable feed. This helped decide how to present content within the application and identifies which navigation and display format participants prefer, independently of the channel they were using.

Activity 3: Build Your Own Screen. This activity comprised a blank phone screen print-out accompanied by standard phone icons and potential features, for participants to design their imagined home screen for the app. This activity was removed after Day 1: it required the longest time to complete of all workshop tasks, and facilitator observations noted that participants placed all the provided design cards onto the layout template without specific sorting or prioritization, largely driven by fatigue and the broad scope of the open-ended task, which induced a blank page problem. The activity yielded limited differentiating data regarding feature priority or layout preferences, and field observations recorded a decrease in participant engagement during subsequent workshop

tasks, so it was removed for the remaining days to shorten the workshop length.

Activity 4: Feature Placement. Participants sorted 14 functional feature cards into three zones: inside the AMEN application, on social media channels, or both (full card list in Supplementary Information, Appendix D). This directly addresses the question of which features must remain in a dedicated application versus those better suited for existing social media channels, directly from the users.

Activity 5: Community and Sharing Map. Using a phone in the centre of the page and design cards representing community members and available digital tools, participants mapped out how they wanted maternal health information to flow to those around them. Participants were encouraged to draw connection lines or barriers between design cards and annotate these lines with their interpretations of how information would flow between community members. This provided an idea of their preferences for community sharing infrastructure, privacy, and sources of maternal information.

3.9 Field Notes and Audio Recordings

Across all three sites and participant groups, field notes were taken by the researchers. These notes captured navigation behaviours, group dynamics, facilitation decisions, unexpected participant responses, and contextual observations that could not be captured by structured instruments. Audio recordings of all sessions were taken with participant consent and subsequently transcribed for thematic analysis. Recording sessions alongside field notes was intended to provide some reliability check on the qualitative data, allowing facilitator observations to be cross-referenced against what participants said, though recording quality varied across sites and this was not always possible in practice. These sources provide qualitative depth and contextual grounding that contextualizes patterns observed in the interaction parts.

3.10 Data Analysis

All data sources, Likert-scale responses, open-ended question responses, co-design artefacts, audio recording transcripts, and field notes, were analyzed using Braun and Clarke's reflexive thematic analysis [3, 4]. Data was first familiarized through repeated reading, then coded inductively across all sources simultaneously; codes were then grouped into candidate themes and reviewed against the full dataset before being defined, named, and written up. NVivo and Excel were used to manage and organize the coding process across the large volume of data. Triangulation across all five sources was vital to the analytical approach: where a theme was supported by multiple sources simultaneously, it was treated as a robust finding, and where evidence appeared in only one source, it was noted but treated with more caution, positioned as a direction for future investigation rather than a confirmed finding.

4 Findings

4.1 Task Performance and Interaction Observations

This section reports what facilitators directly observed as participants worked through the sessions.

4.1.1 Set-Up and Hardware Challenges. Set-up and hardware issues came up across all groups. On Day 1, the app (.APK) was shared via a QR code over a local hotspot, which caused onboarding friction for older CHWs and decision-makers, who struggled to connect to WiFi, enter passwords, and scan the code on low-spec cameras. Facilitators moved to USB sideloading and Quick Share instead. This was compounded by hardware realities: some participants, mainly grandmothers, used non-smartphone devices, and storage space was often limited. Even so, the desire to have the app persisted; one pregnant woman offered to delete her personal photos to make room for it.

4.1.2 Navigation and On-Screen Interactions. **Trial-and-error icon navigation over reading** was common across all three channels. On Day 1, a pregnant woman clicked through unrelated AMEN menus, searching emergency contacts and her profile while looking for a recipe in the clearly labeled "Food" section. The same pattern appeared on YouTube: despite bilingual titles, pregnant women often clicked video cards without reading them. One mother only found the right video by recognizing audio from a nearby participant's phone; another bypassed titles entirely and asked to see a thumbnail so she could match it visually.

Health workers echoed this on Day 3, saying they "don't want to read everything" and preferred familiar icons. Grandmothers summarized the same preference as "No too much charra, charra!" (no too much chattering/reading). This preference shaped language choice too: most health workers and pregnant women chose English for the app interface despite Sepedi being their first language, with one explaining, "we speak Sepedi... but reading English is easier to make out what it is saying."

Clickable element blind-spots were a recurring barrier. Facilitators repeatedly saw participants fail to recognize text lists or cards as interactive; groups of health workers and grandmothers both reached the recipe list and assumed the task was done, needing prompting to actually tap an item. The same happened during onboarding: after selecting a language, participants stared at the screen without realizing they had to tap "Continue," and didn't realize a name field needed to be tapped to bring up the keyboard.

Directory confusion and back-button friction disoriented users across all platforms. AMEN's folder structure caused real navigation errors. On Day 3, a CHW navigated into the trimester sub-folders but couldn't find her way back to the home page, and finding audio files took longer than expected due to nested folders. This extended to social media: participants struggled to return to the AMEN profile after watching a video on TikTok or YouTube, often exiting the app entirely, with YouTube's autoplay making this worse. Small back buttons and a small bottom navigation bar made this harder still; grandmothers on Day 2 needed their fingers guided to the button. The clearest failure was the Emergency page: despite a bright red icon labeled "Emergency," participants across

all days missed it completely, at one point searching for emergency numbers inside the nutrition folder.

4.2 Inter-Generational Taboo Resistance

A pattern of quiet defiance around dietary taboos emerged across all three sites. Pregnant women reported restrictions enforced by grandmothers: no boiled eggs (believed to prolong labor), no oranges (believed to cause neonatal jaundice), but described working around these rather than confronting them. One woman always waits until her grandmother leaves the house to cook eggs, cleaning up before she returns; another said, "I like eggs, but the grannies say we must not eat eggs, but I cheat." Participants felt changing their grandmothers' minds was not possible given their "old mentality," so they kept their choices private.

CHWs identified a similar dynamic with traditional healers: parents often delayed clinic visits and missed immunizations, treating sick children with herbal remedies first and only presenting at the clinic once illness became severe.

4.3 Likert Scale Survey: Channel Comparisons

Scores are on an adapted 1 to 3 scale (1 = Disagree, 2 = Neutral, 3 = Agree).

Across all 51 participants, AMEN scored highest overall (25.90/30), narrowly ahead of YouTube (25.06) and TikTok (24.63). By group, AMEN scored highest among health workers and decision-makers, but lowest among pregnant women, who favored YouTube and TikTok (full breakdown in Appendix E). These lower usability and findability scores directly mirrored the trial-and-error icon navigation and nested folder confusion observed during session tasks.

Usability Dimension	AMEN	YouTube	TikTok
Trust	2.96	2.89	2.86
Daily Value	2.90	2.74	2.84
Clarity	2.84	2.76	2.68
Enjoyment	2.81	2.66	2.67
Reuse	2.82	2.65	2.79
Cultural Fit	2.74	2.77	2.77
Look & Feel	2.68	2.70	2.69
Findability	2.52	2.33	2.66
Navigation	2.48	2.31	2.41
Independent Use	2.36	2.42	2.40

Table 1: Balanced Overall Mean Usability Scores (1–3 Likert Scale) Across Ten Dimensions by Channel (Normalized Across Stakeholder Groups)

Navigation, findability, and independent use scored lowest for AMEN, directly matching the icon-navigation and back-button struggles noted above.

4.4 Open-Ended Responses

What did you find most useful? (OE1) Health workers valued home gardening's practical and health benefits: "I found it very important to have a vegetable garden at home because you can benefit from it... you can get exercise by working in your vegetable garden." Pregnant women found the app's layout more complex than social media. Decision-makers appreciated the traditional food guidance and recipes, finding the app "straightforward" next to searching social media.

Table 2: Mean Channel Preference Scores by Participant Group (Scale 1–3)

Participant Group	Sample (*n*)	AMEN App	TikTok	YouTube
Health Workers	21	2.76	1.90	1.33
Decision-Makers	16	2.33	2.40	1.27
Pregnant Women	14	1.86	2.36	1.79
Balanced Overall Mean	**50**	**2.32**	**2.22**	**1.46**

What was harder on the app than on YouTube/TikTok? (OE2) Health workers found YouTube slow to navigate at first, though it got easier with support. Pregnant women found the app confusing simply because it was unfamiliar, unlike TikTok; one noted, "the audio part was hard to find on the app but it was easier to find it on the YouTube and TikTok because the app was too much confusing." Decision-makers struggled with small text and physical visibility, wanted emergency contacts prioritized, and found searching for pregnancy-safe foods on YouTube difficult since they weren't familiar with the platform.

What would stop you from following health advice? (OE4) Health workers pointed to unemployment, poverty, and busy schedules as structural barriers. Pregnant women named data costs as their main restriction, valuing AMEN's offline access directly: "the app does not require money, and it is able to help me even when I don't have data." Decision-makers raised financial constraints tied to state grants, alongside traditional practices and maternal authority that can override app guidance.

Would you share this with family/community? (OE5) Health workers were confident their communities would take up health guidance, suggesting WhatsApp and TikTok for spreading it. Pregnant women trusted their partners' support and saw TikTok as the best sharing channel due to its reach. Decision-makers believed families would listen, but emphasized enforcing traditional standards, noting expectations like not exposing the belly, keeping the unborn child warm, avoiding cold water, smoking, and alcohol.

4.5 Co-Design Artifact Results

Scores for Activities 1 and 2 are normalized as mean preference values per participant (n=21 Health Workers, n=16 Decision-Makers, n=13 Pregnant Women).

Pregnant women favored TikTok's familiar swiping ("on tiktok can go through without having to struggle"), but ranked AMEN first when it was chosen, valuing its focused pregnancy content ("has all the pregnant woman needs to know"). Health workers split between data efficiency, ease of use, and wanting to build awareness for the newer AMEN app, occasionally ranking TikTok first for its reach ("most people use it"). Across groups, low rankings often reflected unfamiliarity rather than dislike, as one health worker noted about YouTube: "also difficult because it was my first time use YouTube but later I understand it" (based on a small subsample of 8 justifications). This ranking matches the Likert pattern: AMEN and TikTok closely preferred, YouTube trailing.

Table 3: Mean UI Layout Preference Scores by Participant Group (Scale 1–3)

Participant Group	Sample (n)	Folders (Directory)	Swipe (Feed)	Search Bar (Direct)
Health Workers	21	2.19	2.00	1.81
Decision-Makers	16	2.00	2.27	1.73
Pregnant Women	14	2.00	1.71	2.29
Balanced Overall Mean	**50**	**2.06**	**1.99**	**1.94**

Pregnant women favored the search bar, wanting to "find what you want easily" without digging through folders, which they called "the hardest because people are lazy to read and... it needs someone to come to the files one by one." Health workers preferred folders for organization; debate between search and swipe led some to open WhatsApp and TikTok to compare formats directly, with one arguing "searching is faster than scrolling... if you scroll, I can go to the toilet and find you still" looking for a video. Decision-makers preferred the swipe feed for its simplicity, needing no reading or folder navigation (based on a small subsample of 4 justifications).

4.5.1 Activity 3: Build Your Own Screen. This activity was dropped after Day 1, so findings are limited to Evelyn Lekganyane. A split emerged: pregnant women designed simple, warm welcome screens with little educational content, while health workers and decision-makers built denser layouts around modules like "Foods to Eat for a Strong Baby," "Recipes," and "Clinic Visit Prep." All groups prioritized emergency contacts, peer support chats, clinic reminders, a calendar, and the profile menu.

4.5.2 Activity 4: Feature Placement. Core educational modules landed almost entirely on **AMEN**: "what foods to eat for a strong baby," "how to support the new mom," "how to grow your own food at home," "what to do when you have strong cravings." The **"Both"** zone held clinic-related features ("clinic visit reminders," "what to do before and during your clinic visit") and myth-busting/food-safety content. **Social media** took features tied to sharing and habitual use: sharing with other pregnant women and family, daily reminders, and week-by-week tracking. The most commonly **unplaced** cards were the two sharing features, suggesting hesitation around sharing pregnancy content publicly. These findings show that participants intuitively understood that no single tool does it all and that they believe AMEN is for secure, clinical, offline safety, whereas social media is for discovery and reach.

4.5.3 Activity 5: Community and Sharing Map. 45 completed sheets were collected, with qualitative notes from 22 participants.

Maps took different shapes: grids, proximity rings, or networks radiating from the phone at the center. Grid layouts often paired roles with function; a health worker linked the father card to "to be prepared for transportation," echoed by a decision-maker's "to buy good food." Proximity rings placed the most important roles closest to the phone, pushing others, like the traditional healer, to the furthest corner, labeled "least important."

Participants assigned content-creation roles across the community: CHWs as professional creators giving health talks, grandmothers and mothers modeling appropriate pregnancy behavior on video, and community leaders and traditional healers documenting traditional care practices. Pregnant women were seen as the main audience, needing content to "know about their pregnancy and what they should eat." Several noted a preference for audio-visual content, since "listening is better than in all aspects."

Card	Frequency	% of Maps
Pregnant Women	34	75.6%
Community Health Worker	32	71.1%
Community	28	62.2%
Nurse	27	60.0%
Pregnancy + App	27	60.0%
Other Caregivers	26	57.8%
Mother	26	57.8%
Social Media	24	53.3%
Mobile App with Educational Videos	24	53.3%
Mobile Pregnancy/Baby Trackers	24	53.3%
WiFi Connected to Internet	23	51.1%
Father	22	48.9%
Mobile Data	20	44.4%
Community Leaders	19	42.2%
Traditional Healers	16	35.6%
Chat Bot	15	33.3%

Table 4: Card Placement Frequency Across Completed Maps for Activity 5 ($n = 45$).

Connectivity shaped these maps too: "mobile data" and "WiFi connected" cards were linked to social media and chatbots, while AMEN was repeatedly annotated as not needing WiFi. Platforms mapped to distinct roles: WhatsApp for family sharing, Facebook linking community to pregnant women, TikTok for sharing with friends, and YouTube as the platform "to share with all people."

4.6 Content Priorities and Preferences

Across the co-design activities, participants converged on which content mattered most and how they wanted it delivered.

Priority content areas. Across Activities 4 and 5, participants prioritized nine content areas: maternal nutrition, clinic readiness, week-by-week education, family support, traditional rules, peer storytelling, budget meals, father modules, and traditional healer topics (full details in Appendix F). The peer-storytelling format grew directly out of the farming video, which participants liked for featuring a relatable peer rather than an outside expert. The myth-busting module reflects a clear desire to separate fact from fiction without dismissing local customs.

Format preferences. Video was consistently preferred over text, but its data cost was a recognized barrier. Pregnant women proposed a workaround: a text-light summary slide at the end of each video that could be screenshotted, giving a near-zero-storage, zero-data reference without needing to re-watch or re-download. Familiar local background music in the slideshows was also noted as making content feel more engaging and culturally appropriate.

Who should create content. Participants distributed content-creation responsibility across community roles rather than expecting it only from researchers: CHWs delivering health talks, grandmothers and mothers modeling appropriate pregnancy behavior on video, community leaders and traditional healers documenting traditional care practices, and researchers or clinic staff providing the clinical content participants trusted most.

5 Discussion

This study set out to determine which digital channel best serves pregnant women and their support structures in rural Limpopo for maternal health content delivery. The findings suggest this was the wrong question: while AMEN scored highest overall, no single platform met the full range of digital literacy, access constraints, and communication preferences across this community. Therefore the most substantiated conclusion is that effective delivery here requires a multi-channel ecosystem, with each platform playing a distinct, complementary role rather than competing to be the single solution.

5.1 The Multi-Channel Ecosystem

Much of the current maternal health literature assumes a dedicated mobile app is the best way to deliver health information. Our findings don't support this: different groups within the same community have different digital habits, literacy levels, and access constraints that no single platform can meet. Across all three stakeholder groups, participants treated social media platforms like YouTube, TikTok, and WhatsApp as primary, familiar channels alongside AMEN, with the answer depending heavily on which stakeholder group is asked.

This is clearest in the mismatch between pregnant women and health workers. Pregnant women's Likert scores and rankings favored social media, driven by familiarity, but they also named data costs as a barrier and valued AMEN's offline access, pointing to a hybrid model where social media is the accessible entry point and the app the affordable fallback. This mirrors earlier findings that rural users prefer audio and video content but are excluded from richer apps by high data costs and poor connectivity [13]; our data shows this trade-off playing out directly in participants' own justifications, not just as an access barrier but as a stated design preference. Health workers, by contrast, strongly favored AMEN on both measures, trusting it as a dedicated, offline information tool, while still valuing social media's reach for driving initial awareness. This contrast is notable next to MomConnect, which achieved large-scale adoption but was limited in its interactivity by its reliance on USSD and WhatsApp [2, 14]; our findings suggest that scale and interactivity may simply require different channels altogether, rather than one platform solving for both.

Decision-makers showed a further contradiction: they rated AMEN highest on the Likert scale, including moderate-to-positive Navigation and Independent Use scores (2.40 and 2.06), despite facilitators repeatedly having to guide them through basic tasks, and none of the grandmothers having brought reading glasses. This suggests some verbal agreement bias, rating the app well out of appreciation rather than ease of use. The bias didn't erase all differences, though: decision-makers still rated TikTok's Navigation and Independent Use notably lower (1.75 and 1.69) despite similar support on both platforms. Their channel ranking also inverted their Likert preference, placing TikTok (2.40) just above AMEN (2.33), matching their layout preference for the low-text swipe feed (2.27) over folders (2.00) or search (1.73), echoing the "charra, charra" reading fatigue noted earlier. This points to pairing a private, zero-data clinical app with low-text visual content, like short videos or voice notes, as a realistic route for older decision-makers, without assuming TikTok alone is easy for them to use.

The Feature Placement results support this directly: participants placed core education on AMEN, habitual and high-reach content on social media, and clinic/myth-busting content in "both," treating the platforms as functionally distinct rather than competing. The frequent non-placement of sharing features suggests privacy sensitivities around pregnancy content, favoring private channels like WhatsApp or private TikTok circles over public posts. The Community and Sharing Map reinforced this: participants consistently linked AMEN to offline, static use and social media to the connected layer that drives traffic toward it.

This pattern extends beyond platforms to authority itself. Rather than directly challenging traditional dietary rules, elders or traditional healers, participants quietly worked around them. They preserved family harmony while still acting on their own informed preferences, reflected in traditional healers being placed furthest from the center and labeled "least important" on the sharing maps. This adds nuance to prior work describing tension between cultural practice and evidence-based dietary guidance in this context [5, 15]: our findings suggest the tension is rarely expressed as open conflict, but as quiet, private negotiation. Interventions that force visible contradiction with elders are likely to fail; private, low-visibility tools like AMEN's offline, personal format fit this dynamic better, and align with earlier findings that trust in digital health content builds gradually through relatable stories and peer testimonials rather than confronting existing authority head-on [8, 9]. The resulting ecosystem is cooperative rather than competitive: the offline AMEN app serves as a secure, data-free database of clinical facts; TikTok and YouTube provide visual familiarity and wider reach; and WhatsApp bridges the gap through low-cost, trusted peer-to-peer sharing.

5.2 The Trust-Usability Gap

A gap was also found between trust and usability. AMEN scored highest on trust (2.96/3.00) but lowest on findability (2.52), navigation (2.48), and independent use (2.36): participants valued the content but struggled to use the interface, consistent with the small buttons, interactive blind-spots, and nested folders observed throughout the sessions. This mirrors a similar gap identified in a South African user-centred redesign, where addressing usability issues directly raised initial adoption from 41.2% to 72.4% and 90-day retention from 29.7% to 51.8% [12], suggesting our redesign recommendations below are working from a similar, well-supported premise. The following recommendations follow directly from those observations and are meant to generalize beyond AMEN.

A hybrid layout and scrollable feed. Layout preferences split by group: folders for health workers, search for pregnant women, swipe feed for decision-makers. A combined design could offer folders and search on the home screen, with a full-screen scrollable feed once inside a folder, plus a large exit button to prevent users getting lost mid-scroll.

Visual-first navigation and better iconography. Given the strong preference for icons over text, labels should stay short, with a text-size adjuster for users without reading glasses. The Emergency page's failure shows generic icons aren't enough; icons should draw

on symbols familiar to the Limpopo context, such as an ambulance or clinic building, ideally validated through further community research.

Fixing interactive blind-spots and adding voice guidance. Buttons and cards should use raised or shadowed styles to look tappable. Onboarding should include optional spoken guidance in Sepedi and English, and a read-aloud feature on text blocks, matching participants' stated preference that "listening is better in all aspects."

Role-based modes and a father role. The screen-building activity showed different design instincts by group, suggesting a clinical "Tool Mode" for health workers and a warmer, low-text "Companion Mode" for pregnant women. Since fathers emerged as key economic and logistical supporters in the sharing maps, a dedicated father's mode covering budgeting, gardening, and emergency transport checklists is also worth exploring.

A personalized chatbot. Pregnant women independently sketched chatbot features and referenced them again in the sharing exercise. A future text/voice chatbot, restricted to a clinically verified content database, could let mothers ask questions directly in Sepedi and get trusted, personalized answers.

Low-data sharing. Since data costs were pregnant women's most cited barrier, and they valued AMEN specifically for being data-free, the app should support offline sharing options like text-based recipe copying, low-data PDF export via WhatsApp, and peer-to-peer installs via Bluetooth or Quick Share.

These are grounded in how participants actually interacted with the screens, and represent validated recommendations even though the redesigned interface itself hasn't yet been independently tested, a gap addressed in Future Work below. Our set-up and navigation observations, participants confidently engaging with a working AMEN prototype despite hardware friction, suggest that any future evaluation of these redesign recommendations should likewise use working prototypes rather than static mockups.

5.3 Methodological Contributions

This study offers lessons for participatory co-design in rural, low-literacy contexts.

Sessions ran longer than planned, and participant numbers shifted (51 of a planned 54) due to antenatal check-ups, transport issues, and higher-than-expected CHW turnout. Activity 3 was dropped after Day 1 due to fatigue and low engagement. Splitting the workshop into Part A (task evaluation) and Part B (co-design) mattered: without hands-on practice with the actual prototypes first, participants would have lacked the vocabulary to design meaningfully afterward. The three channels effectively worked as probes, giving participants a concrete baseline for informed decisions in the ranking and feature-placement tasks, letting them test features against realistic goals rather than facing abstract, step-by-step instructions.

Groups needed different levels of support; pregnant women were generally younger and more digitally confident, while grandmothers and health workers needed constant facilitation. Paper worksheets also proved limited for capturing reasoning in this setting: 50 participants completed the card-sorting and ranking tasks, but only a small subset wrote out justifications, so facilitators relied heavily on audio recordings and field notes instead. The abstract community-mapping task was hard to explain verbally on the first two days; switching to a worked example using an unrelated scenario on the final day let participants grasp it immediately, producing some of the richest co-design data collected. This suggests that abstract brainstorming creates a "blank page problem" in low-literacy settings that extends beyond physical materials to how abstract tasks are explained verbally, not just what tools are handed to participants.

Agreement bias, participants defaulting to top ratings out of politeness, was a recognized risk that several design choices likely compounded: the 3-point scale's known reliability limit [10], verbal administration adding interviewer presence as social pressure, and facilitators writing answers for participants without reading glasses, meaning some responses weren't fully independent to start with. Decision-makers' pre-existing relationship with the UL team may also have added a stronger polite bias toward high scores than the newly recruited pregnant women, who gave more direct, critical feedback. Even so, the data suggests this bias inflated scores without erasing real differences: four participants gave a uniform 30/30 across channels, and 28 gave at least one channel a top score, but the remaining scores were more varied and deliberate, isolating genuine friction points like navigation and layout, consistent with the decision-makers still rating TikTok's Navigation and Independent Use lower than AMEN's despite similar support. Combined with the forced-choice rankings and qualitative justifications, this suggests the dataset reflects authentic, if inflated, user perception rather than uniform positivity, and that combining quantitative task data with qualitative feedback gives a more complete picture of usability in these settings.

Taken together, these lessons suggest that working prototypes as design probes, structured task evaluation kept separate from creative co-design, and a reliance on transcribed discussion over written worksheets are important design choices for participatory research in similar contexts. This aligns with earlier calls for co-design in rural, multilingual settings to be led by native speakers rather than relying on automated translation [6], since much of what worked here, from facilitator-led scribing to verbal scenario walkthroughs, depended on native Sepedi speakers being present throughout. Agreement bias in Likert surveys should be treated as a structural feature of low-literacy evaluation rather than a data quality failure; recognizing this contributes to more honest, robust HCI studies in similar contexts.

5.4 Future Work

Four directions follow from this study. First, dedicated research into iconography and imagery is needed to ensure the application feels familiar to users. Second, a Phase 2 evaluation of the redesigned AMEN application should measure whether these structural changes improve independent usability. Third, a formal comparative evaluation of WhatsApp as a primary content channel is warranted, given participants' consistent preference for it in daily sharing. Finally, a longitudinal study is needed to assess whether this multi-channel ecosystem produces lasting change in household maternal nutrition habits.

5.5 Limitations

This study has several limitations worth noting.

The 3-point Likert scale, chosen to reduce cognitive load, carries a known reliability limit [10], and verbal administration, while removing a literacy barrier, likely contributed to some agreement bias. The data suggests this inflated scores rather than erasing real differences between channels, but it remains a limitation of the survey data.

The final sample of 51 participants (against a planned 54) was shaped by antenatal check-ups, transport constraints, and unexpectedly high CHW interest, resulting in uneven group sizes across sites. Methodological adjustments during fieldwork also limited certain data sources: Activity 3 was dropped after Day 1 due to participant fatigue, restricting its findings to Evelyn Lekganyane Clinic, while qualitative insights for Activity 5 are primarily drawn from Day 3, as initial facilitation challenges in communicating the abstract task on Days 1 and 2 were only resolved after introducing an unrelated visual example.

Researcher positionality is a further limitation on how the data was collected and interpreted, and is discussed in full in Section 3.2.

Finally, the redesign recommendations here are grounded in fieldwork data but have not yet been tested as an implemented interface, and the study itself is bound to three clinics in the Capricorn District of Limpopo, leaving open how far the proposed multi-channel model generalizes elsewhere.

6 Conclusion

This study set out to identify the best digital channel for delivering maternal nutrition content in rural Limpopo, and found that no single channel fits the bill. Pregnant women, health workers, and decision-makers each brought different digital habits, trust patterns, and access constraints to the fieldwork, and AMEN, TikTok, and YouTube each met a different part of those needs rather than one meeting all of them. The most useful contribution here is not a single winning platform, but a model for how a dedicated app and existing social media can work together, alongside a set of concrete, transferable design recommendations for closing the gap between trust and usability in similar low-resource, rural contexts.

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Supplementary Information

Appendix A: Predefined Task Interaction — Full Task Wording

Five identical scenario-based tasks were administered across the AMEN application, TikTok, and YouTube channels (see Section 3.8.1). Each task was read aloud in both English and Sepedi, and phrased differently depending on the participant's stakeholder group (Pregnant Woman, CHW, or Family/Decision-Maker), while targeting the same underlying navigation goal.

(1) **Finding a recipe.** *Pregnant Woman:* "You are planning supper tonight. Can you find a healthy meal you could cook for yourself using ingredients you might have at home?" (*Mosadi wa Moimana:* "O rulaganya dijo tša mantšiboa lehono. Na o ka hwetša dijo tša maphelo tšeo o ka iphekelago tšona o diriša dijo tšeo mohlomongwe di lego ka gae?")
OR CHW/Family: "The pregnant woman you care for needs a recipe for a healthy meal for supper. Can you find her a recipe for a meal using the ingredients she may have at home?"

(2) **Home gardening video.** *Pregnant Woman:* "A friend of yours started her own home garden to grow fresh vegetables, and now you want to start one too. Can you find and watch a video that could help you learn how to grow your own healthy food in your garden?"
OR CHW: "You want to find a video you could show to a pregnant woman you support, to help her learn how to grow her own healthy food in a home garden. Can you find and watch a video that would be useful to show her?"
OR Family: "Someone in your family is pregnant, and you want to start a home garden to grow fresh, healthy food for your family and for her. Can you find and watch a video that could help you learn how to grow your own food?"

(3) **Trimester-specific nutrition download.** *Pregnant Woman:* "You are in your 2nd trimester (~Weeks 14–27) and want to know what you should be eating at this stage of your pregnancy. Can you save or download something that gives you meal or food ideas for where you are now?"
OR CHW/Family: "You are supporting a pregnant woman who is in her 2nd trimester (~Weeks 14–27). Can you save or download something that gives you meal or food ideas you could recommend to her for where she is now in her pregnancy?"

(4) **Finding an audio message.** *Pregnant Woman:* "You want to learn more about nutrition during pregnancy, but you are tired and do not feel like reading. Can you find an audio message to listen to instead?"
OR CHW: "You are interested in growing your knowledge about nutrition during pregnancy but are tired and do not want to read. Can you find an audio message to listen to instead?"
OR Family: "You want to learn more about nutrition during pregnancy together with your family. Can you find an audio message to play out loud so you can all learn?"

(5) **Locating emergency numbers.** *Pregnant Woman:* "You are feeling a sudden pain in your stomach and need to get to the hospital urgently. Can you find the emergency numbers to call for help?" (*Mosadi wa Moimana:* "O kwa bohloko bjo bogolo ka tšhoganetšo ka mpeng gomme o swanetše go fihla sepetlele ka pela. Na o ka hwetša dinomoro tša tšhoganetšo tšeo o ka letšetšago go hwetša thušo?")
OR CHW/Family: "A pregnant woman you are supporting is experiencing a sudden pain and needs to get to the hospital urgently. Can you find the emergency contact numbers quickly so you can get her the help she needs?"

Appendix B: Likert Usability and Satisfaction Scale — Full Item List

A 10-item scale scored as 1 (Disagree), 2 (Neutral), and 3 (Agree) was administered for each channel (see Section 3.8.1), read aloud in Sepedi with the English translation given to the participant.

Appendix C: Semi-Structured Open-Ended Question Materials

Individual semi-structured questions followed the channel interaction and surveys, structured around three focus areas: comparative usability, digital familiarity, and socio-economic constraints (see Section 3.8.1). Questions were written and read aloud in Sepedi and English.

(1) **(OE1)** What did you find most useful or important in the app?
(2) **(OE2)** Was there anything difficult or confusing to find, access, or use in the app that was easier to do on YouTube or TikTok?
(3) **(OE3)** Are there other apps or digital platforms you are familiar with and use regularly on your phone?
(4) **(OE4)** What barriers or restrictions would make it difficult or prevent you from following health advice or sharing information?
(5) **(OE5)** Will it be easy to share this information with your family or community, and do you think they will listen and support you?

Note: OE3 is included here for completeness of the instrument even though its findings were folded out of the main Results text — the actual questions administered should always be documented in full here regardless of what made it into the main paper.

Appendix D: Co-Design Session Materials

The co-design session made use of low-fidelity physical materials, including cards, templates, and art supplies (see Section 3.8.2).

General materials: All co-design activities used physical, non-digital materials so no screens were required during the session: pre-printed channel preference cards (Activity 1); pre-printed content-layout option cards representing a folder structure, a search/conversational interface, and a scrollable feed (Activity 2); a blank phone-screen print-out with pre-cut icon/feature stickers and pens (Activity 3); pre-printed feature cards sorted across an "App" / "Social Media" / "Both" zone layout (Activity 4); and an A3 sheet pre-printed with a phone drawn in the centre, along with pens, coloured markers, and small sticker dots or sticky notes used to represent people and connections (Activity 5).

Dimension	Item Statement (Sepedi / English)
Trust	Ke dumela gore tshedimošo ya maphelo yeo e lego mo kanaleng ye e nepagetše ebile e a botega (I believe the health information on this channel is correct and trustworthy)
Daily Value	Se se mphile selo sa mohola seo nka se dirišago bophelong bja ka bja letšatši le letšatši (This gave me something I could use in my daily life)
Clarity	Tsela yeo diteng di bego di bontšhwa ka yona e be e kwagala gabotse ebile e le bonolo go e kwešiša (The way the content was displayed was clear and easy to understand)
Enjoyment	Kanale ye e be e le bonolo ebile e thabiša go e diriša (This channel was fun/enjoyable to use)
Reuse	Nka diriša kanale ye gape go hwetša dikeletšo tša boimana (I would use this channel again to get tips for the pregnancy)
Cultural Fit	Diteng di be di bonagala di hlametšwe motho wa go swana le nna le setšhaba sa gešo (The content felt like it was made for someone like me and my community)
Look & Feel	Ke ratile tsela yeo kanale ye e bonagalago ka yona le kamoo go bego go le ka gona go e diriša (I liked the way this channel looked and felt to use)
Findability	Ke hweditše go le bonolo go hwetša seo ke bego ke se nyaka (I found it easy to find what I was looking for)
Navigation	Ke be ke tseba moo nka yago gona ka morago ntle le go gakanega (I knew where to go next without getting confused)
Independent Use	Nka kgona go diriša se ka noši ntle le thušo ya motho yo mongwe (I would be able to use this on my own without anyone helping me)

Table 5: Full Likert Usability and Satisfaction Scale (1–3 scale) administered per channel

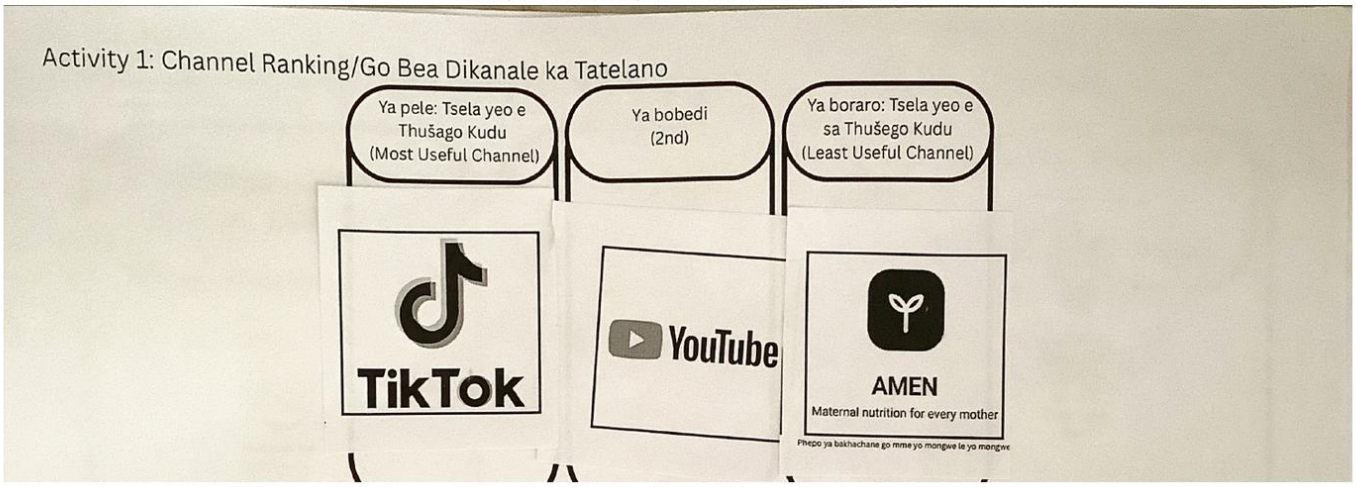
Activity worksheets: The five activities are reproduced below, one landscape page per activity, each showing the blank template administered to participants above a completed participant example (drawn from worksheet D-D-02). Activity 3 has no completed example since it was discontinued after Day 1 (see Section 5.3); the blank template is shown on its own.

Activity 1: Channel Ranking / Go Bea Dikanale ka Tatelano

Blank template (administered to participants)

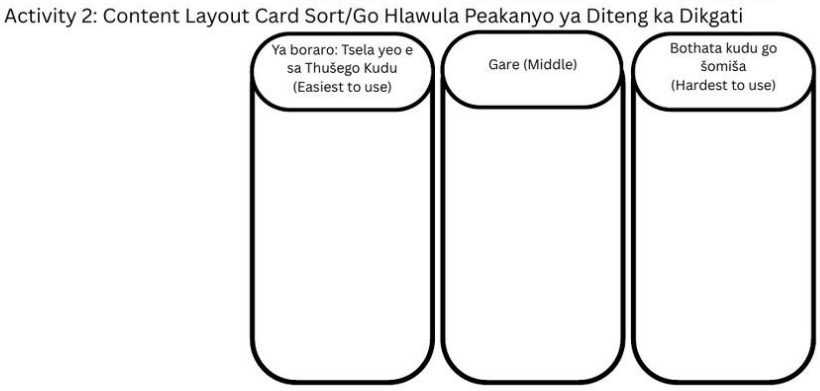


Completed example — worksheet D-D-02

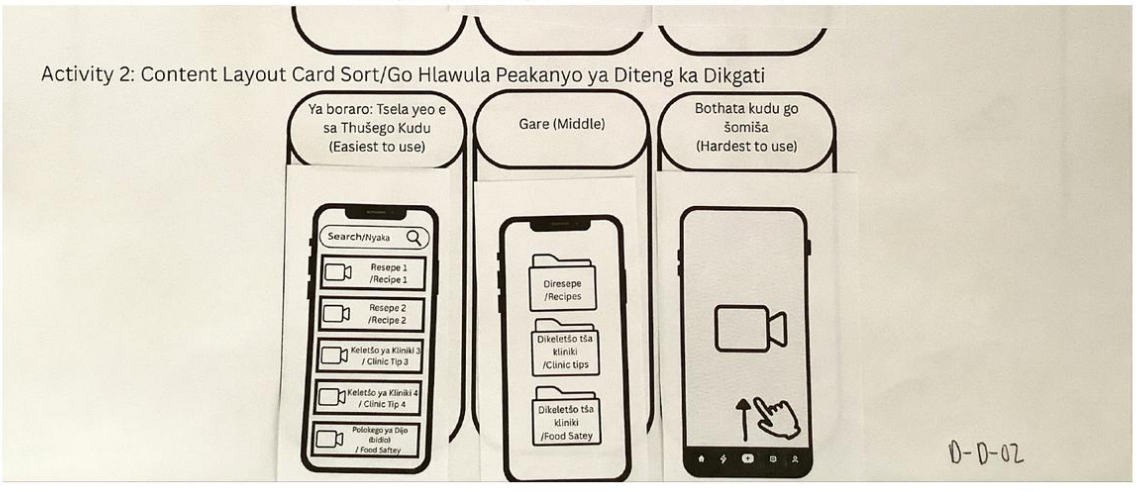


Activity 2: Content Layout Card Sort / Go Hlawula Peakanyo ya Diteng ka Dikgati

Blank template (administered to participants)



Completed example — worksheet D-D-02



Activity 3: Build Your Own Screen / Itirele Sekirini sa Gago

Blank template (administered to participants, Day 1 only)

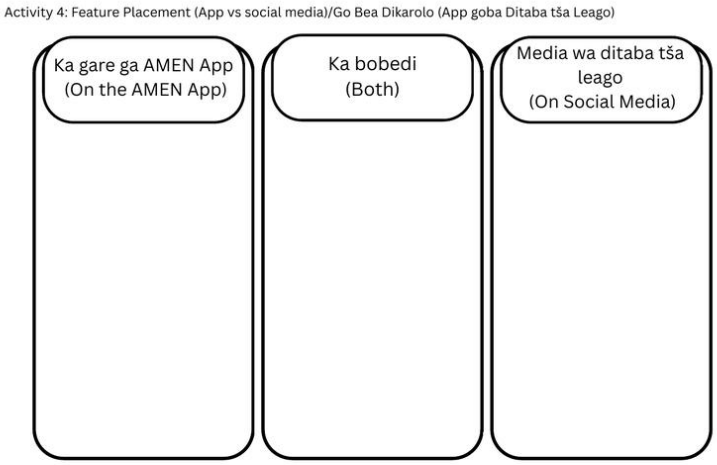
Activity 3: Build Your Own Screen/Itirele Sekirini sa Gago



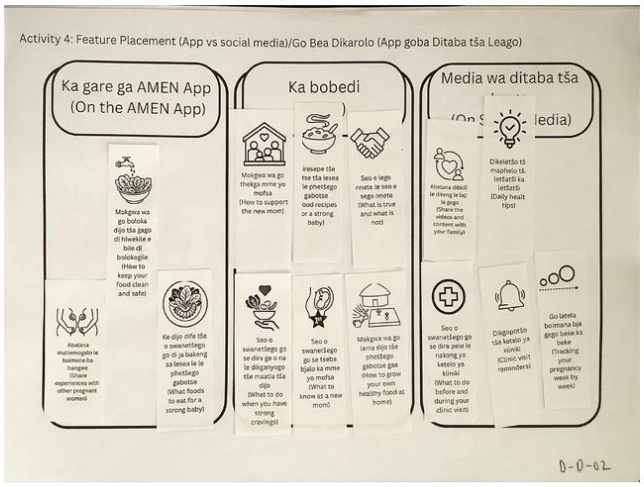
No completed example available — Activity 3 was discontinued after Day 1 (see §3.8.2).

Activity 4: Feature Placement / Go Bea Dikarolo (App goba Ditaba tša Leago)

Blank template (administered to participants)



Completed example — worksheet D-D-02

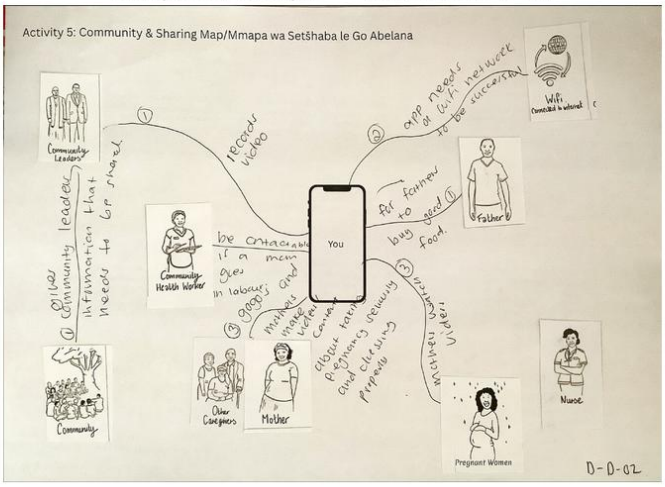


Activity 5: Community & Sharing Map / Mmapa wa Setšhaba le Go Abelana

Blank template (administered to participants)



Completed example — worksheet D-D-02



Activity 4 Feature Card List: Participants sorted **14** functional feature cards into three zones (AMEN application / social media / both). The zone shown for each card below is as reported in the Findings section (§4.5.2); one card was not discussed by zone in that section and is marked accordingly.

- What foods to eat for a strong baby (AMEN)
- How to support the new mom (AMEN)
- How to grow your own healthy food at home (AMEN)
- What to do when you have strong cravings (AMEN)
- Clinic visit reminders (Both)
- What to do before and during your clinic visit (Both)
- What is true and what is not (Both)
- How to keep your food clean and safe (Both)
- Good recipes for a strong baby (Both)
- Daily health tips (Social Media)
- Tracking your pregnancy week-by-week (Social Media)
- Sharing with other pregnant women (unplaced / Social Media)
- Sharing with your family (unplaced / Social Media)
- What to know as a new mom (zone not documented)

Appendix E: Likert Scores Disaggregated by Stakeholder Group

Channel	Participant Group	*N*	Mean Score (/30)
4*AMEN App	Health Workers	21	28.48
	Decision-Makers	16	27.06
	Pregnant Women	14	20.71
	All Participants	**51**	**25.90**
4*YouTube	Health Workers	21	26.67
	Decision-Makers	16	24.63
	Pregnant Women	14	23.14
	All Participants	**51**	**25.06**
4*TikTok	Health Workers	21	25.48
	Decision-Makers	16	23.75
	Pregnant Women	14	24.36
	All Participants	**51**	**24.63**

Table 6: Overall Likert Usability Scores by Channel and Participant Group (Maximum Score: 30) — referenced from Findings §4.3

E.1 Overall Usability Scores by Channel and Participant Group (out of 30).

	AMEN				YouTube				TikTok			
Dimension	**HW**	**DM**	**PW**	**Bal.**	**HW**	**DM**	**PW**	**Bal.**	**HW**	**DM**	**PW**	**Bal.**
Trust	2.95	3.00	2.92	**2.96**	2.76	3.00	2.92	**2.89**	2.95	2.81	2.83	**2.86**
Daily Value	3.00	2.94	2.75	**2.90**	2.81	2.75	2.67	**2.74**	2.95	2.56	3.00	**2.84**
Clarity	3.00	2.94	2.58	**2.84**	2.85	2.75	2.67	**2.76**	2.74	2.56	2.75	**2.68**
Enjoyment	3.00	2.75	2.67	**2.81**	2.76	2.31	2.92	**2.66**	2.84	2.25	2.92	**2.67**
Reuse	2.95	2.69	2.82	**2.82**	2.90	2.56	2.50	**2.65**	2.89	2.56	2.92	**2.79**
Cultural Fit	2.95	2.81	2.45	**2.74**	2.71	3.00	2.58	**2.77**	2.84	2.63	2.83	**2.77**
Look & Feel	2.95	3.00	2.09	**2.68**	2.71	2.81	2.58	**2.70**	2.79	2.63	2.67	**2.69**
Findability	2.76	2.63	2.18	**2.52**	2.52	1.88	2.58	**2.33**	2.74	2.31	2.92	**2.66**
Navigation	2.50	2.40	2.55	**2.48**	2.52	1.81	2.58	**2.31**	2.74	1.75	2.75	**2.41**
Independent Use	2.67	2.06	2.36	**2.36**	2.50	1.75	3.00	**2.42**	2.68	1.69	2.83	**2.40**

Table 7: Mean Usability Scores (1–3 Likert Scale) Across Ten Dimensions by Channel and Stakeholder Group — referenced from Findings §4.3 and Discussion §5.1. HW = Health Workers ($n = 21$), DM = Decision-Makers ($n = 16$), PW = Pregnant Women ($n = 14$), Bal. = Balanced Overall Mean.

E.2 Mean Usability Scores by Dimension, per Channel and per Group.

Appendix F: Priority Content Bundles — Full Module List

Category	Example Modules / Grounding
Maternal Nutrition & Safety	Foods for a strong baby; recipes; strong cravings; food safety (raw vs. cooked)
Clinic & Emergency Readiness	Clinic visit preparation and reminders; emergency contacts
Pregnancy Education	Week-by-week tracking; "What is true and what is not" myth-busting
Family & Social Support	Supporting the new mom; peer support chats
Traditional Rules & Family Guidelines	Grandmother-expected lifestyle rules (warmth, modesty, avoiding cold water/smoking/alcohol); grandmother- and healer-recorded videos
Peer-to-Peer Storytelling	Local farmer gardening video model, extended to peer testimonials on nutrition and clinic visits
Budget-Friendly Nutrition	Low-cost, locally available ingredients; grant-based food budgeting; garden yield tips
Father-Targeted Modules	Transport planning, food budgeting, practical support checklists
Traditional Healer Content	Limited to documenting safe cultural customs, not clinical advice (reflecting low prioritisation in proximity-ring mapping)

Table 8: Priority Content Bundles Identified Through Co-Design — referenced from Findings §4.6

7 AI and Technological Assistance Declaration

During the preparation of this research paper, generative AI tools and automated editing software were utilized to assist with drafting, structural condensation, formatting, and editorial refinement. The author conceptualized all core research design choices, fieldwork execution, data analysis, and academic interpretations, retaining sole responsibility for the final content.

The specific assistance workflow included the following tools and prompts:

Prompt Engineering & Ideation (Gemini): Gemini was used to construct an expert academic prompt designed to guide subsequent editorial analysis. The prompt established the role, context, and constraints for reviewing draft sections:

"Section by section i will be feeding you my paper please realign to these instructions for each section. Role & Task: You are an expert academic editor specializing in South African public health, human-computer interaction (HCI), and qualitative research methodologies (specifically reflexive thematic analysis). Help me by suggesting edits and ways to condense my Honours research paper draft. Study Context & Key Details: Topic: Evaluating a maternal health content delivery channel (AMEN app prototype, YouTube, and TikTok channels) for pregnant women and support structures in rural Limpopo. My Role: Designed the user interaction plan for testing/codesign; co-facilitated field workshops with research facilitators from the University of Limpopo (UL) and the University of Cape Town (UCT); analyzed qualitative findings using NVivo and Excel via Reflexive Thematic Analysis. Current Issue: The draft is [x] pages over the limit, redundant in places, and may lack tight flow between the methodology, codesign findings, and discussion. Suggestion Priorities: Reduce Word Count & Length (Primary Priority): Aggressively prune fluff, eliminate redundant phrasing, cut structural repetition, and combine overlapping points without losing technical depth or methodological nuance. Coherence & Flow: Ensure smooth transitions between the codesign facilitation phase, the multi-platform evaluation (AMEN vs. YouTube/TikTok), and the thematic analysis findings. Academic Tone & Grammar: Polish the prose to maintain a high-level academic voice appropriate for an HCI/public health paper. Rubric Alignment: Check my writing against the attached rubric criteria."

Structural Editing & Condensation (Claude): Powered by the prompt structure above, Claude was utilized to review draft sections iteratively, identify structural redundancies, suggest prose condensations to adhere to strict page limits, and refine the logical flow between fieldwork findings and the discussion. Claude was also used to format complex tables into clean LaTeX code and assist with structuring appendix sections.

Grammar and Syntax Polish (Grammarly): Grammarly was used for automated grammar checks, spelling corrections, and localized sentence-structure improvements.